Sahaj Bhadja

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Summary

Software engineer with 2+ years building Python services, full-stack workflows, and cloud-backed platforms across FastAPI, Flask, AWS, SQL, CI/CD, and reliable data workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built ETL-style integration pipelines across FastAPI, SQL Server, Supabase, and cloud storage, processing 5K+ monthly jobs with validation checks, reconciliation logic, and CI/CD-backed releases for dependable platform operations.
- Designed and optimized a multi-tenant PostgreSQL schema supporting 200+ users and 10K+ structured records, implementing indexing, query tuning, and validation workflows that reduced latency by 35% in production.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Implemented a production Flask backend on AWS serving 150+ instructors and 10K+ grading requests daily, with CI/CD workflows, secure ingestion, and stable API performance under peak usage.
- Developed React interfaces and backend flows for question upload, grading history, rubric generation, and concurrent evaluation, reducing feedback turnaround by 60% while maintaining dependable service behavior.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer for offline reinforcement learning, improving long-horizon trajectory planning by 18% while developing pipelines processing 100K+ trajectory logs for model evaluation.
- Architected a spatial-temporal encoder integrating map geometry with agent context through multi-head attention, improving closed-loop planning stability by 15% across complex multi-agent scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Engineered cross-platform data-sharing software with microservices and automated validation workflows, using Jenkins, PyTest, and Mockito to strengthen release reliability across Windows, Linux, and macOS environments.
- Troubleshot API integrations with AWS Glue and S3, improving response times by 30% while leading ETL workflow changes that boosted pipeline efficiency 68% and reduced technical debt 20%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN-based resource management platform with REST APIs, WebSockets, and microservices, supporting six workflows and 2M+ synchronized records while reducing code-review turnaround by 25% across releases.
- Automated deployments with Jenkins CI/CD and Docker packaging, achieving 95%+ successful build validation and cutting deployment time by 50% for cloud-backed application updates.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Backend & Fullstack: FastAPI, Flask, Django concepts, REST APIs, React, Vue.js concepts, DRF concepts, WebSockets, microservices, object-oriented design
Cloud & Data: AWS, S3, RDS/PostgreSQL, SQS concepts, Kafka/MSK concepts, Azure SQL, SQL Server, PostgreSQL, Supabase, ETL pipelines, data validation, reconciliation checks, Docker
Delivery: CI/CD, Jenkins, GitHub Actions, PyTest, debugging, monitoring, system design, algorithms, Agile/Scrum, Git

Projects

HMIS Live | *MERN, ABDM APIs, Twilio, Sentry*
- Implemented consent-based patient-history access, appointment alerts, monitoring, and structured API workflows for a healthcare platform, reducing administrative workload by 30% while supporting secure clinician-facing access to records.

Big-Brain Live | *LLaMA-2, FAISS, PostgreSQL, GCP*
- Engineered a cloud ingestion and indexing pipeline processing 50K+ records, implementing incremental updates, metadata persistence, and tuning to reduce retrieval latency by 40% for large document workloads.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Developed and deployed a containerized RESTful data processing API on AWS using CodePipeline and ECS/Fargate, with automated CI/CD releases, validation checks, and reliable recurring processing workflows.